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ZD-HybridNet integrates transformer and LSTM models for explainable zero day cyber attack detection

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Abstract

The exponential growth of interconnected systems has dramatically expanded the attack surface of modern networks, making zero-day attacks, which exploit previously unknown vulnerabilities, a critical threat to organizational security. Traditional signature-based and anomaly detection systems often struggle to identify such previously unseen attack patterns, and while deep learning (DL) offers promising detection capabilities, existing DL-based intrusion detection systems (IDS) suffer from poor generalization to novel attacks and operate as opaque black boxes, lacking interpretability for security analysts. To address these limitations, this paper proposes ZD-HybridNet (Zero-Day Hybrid Attention Network), a novel explainable deep learning framework for large-scale network traffic analysis. ZD-HybridNet integrates a Transformer encoder branch for capturing global feature dependencies and an LSTM branch for modeling sequential traffic patterns, dynamically fused via an adaptive attention-based gating mechanism. This architecture is coupled with an integrated explainability framework that combines model-internal attention and fusion signals with SHAP-based feature attribution to support interpretation of detection decisions. Evaluated on the UNSW-NB15 dataset under a controlled category-holdout protocol in which Shellcode and Worms are entirely excluded from training and validation, ZD-HybridNet achieves an accuracy of 94.04%, a precision of 96.76%, a recall of 93.96%, and an F1-score of 95.34%, outperforming standalone LSTM, Transformer, FT-Transformer, and 1D CNN baselines under the same experimental protocol. Precision-recall, ROC, zero-day-specific, ablation, and SHAP-based analyses further support the model's detection performance and interpretability within the defined Shellcode–Worms holdout setting, while deployment-oriented computational validation remains necessary.

Keywords Zero-day attack detection, Explainable artificial intelligence (XAI), Transformer-LSTM hybrid architecture, Network intrusion detection system (NIDS), Adaptive attention-based fusion, AI-driven secure networking, SHAP feature importance, UNSW-NB15 dataset

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1 Introduction

The exponential growth of interconnected systems and Internet-of-Things (IoT) deployments has dramatically expanded the attack surface of modern networks, creating an environment ripe for sophisticated cyber threats. Among these, zero-day attacks–exploiting previously unknown vulnerabilities–pose an existential risk to organizational security postures, as they can evade conventional signature-based detection and challenge anomaly-based systems [1, 2]. The economic impact is staggering; recent analyses indicate that breaches involving zero-day exploits incur remediation costs 40% higher than average and require nearly 30% longer to identify and contain [3, 4]. This urgency is compounded by the advent of adversarial machine learning, where attackers deliberately craft traffic to evade detection by mimicking benign patterns [5, 6].

In pursuit of robust solutions, the cybersecurity research community has increasingly turned to deep learning (DL) architectures capable of learning intricate, hierarchical representations from raw network traffic. Convolutional Neural Networks (CNNs) have demonstrated efficacy in extracting spatial features from packet-level data [7], while Recurrent Neural Networks (RNNs) and their variants, particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, excel at modeling temporal dependencies across traffic sequences [8]. Recent innovations have explored hybrid models and transformer-based attention mechanisms to capture both spatial and long-range temporal correlations [9, 10]. Despite these advances, two critical limitations persist. First, most DL-based intrusion detection systems (IDS) are trained and evaluated on datasets containing known attack variants, resulting in poor generalization to genuinely novel, zero-day attack methodologies [11, 12]. Second, these complex models typically operate as opaque black boxes, offering no interpretable justification for their predictions. This lack of explainability severely hinders practical deployment in Security Operations Centers (SOCs), where analysts require actionable intelligence to triage alerts and initiate response protocols [13, 14].

While nascent efforts have emerged to address explainability in security contexts, often applying post-hoc techniques like SHAP or LIME [15, 16], and others have proposed specialized architectures for novelty detection [17, 18], a significant research chasm remains. There is a pressing need for an integrated framework that fundamentally couples high-fidelity zero-day detection in large-scale traffic with interpretable model-internal and post-hoc evidence. Such a framework must not only identify novel attacks with high precision but also elucidate the specific traffic features and contextual patterns that trigger the alarm, thereby bridging the gap between algorithmic output and human analyst comprehension.

To address this multifaceted challenge, we propose ZD-HybridNet (Zero-Day Hybrid Attention Network), a novel explainable deep learning framework designed for large-scale network traffic analysis. ZD-HybridNet is architected from the ground up to detect attack categories withheld during model training and validation while providing transparent, human-interpretable rationales for its decisions. The principal contributions of this work are threefold:

- *ZD-HybridNet architecture:* We propose ZD-HybridNet, a novel hybrid deep learning architecture that combines Transformer-based attention mechanisms with LSTM temporal modeling for zero-day attack detection. The model features a

dual-branch design with: (1) a Transformer encoder branch with attention pooling (AttentionPooling class) that captures global dependencies across network features, and (2) an LSTM branch that models sequential patterns in traffic data. These branches are integrated through a learnable fusion gating mechanism (FusionGating class) that dynamically balances their contributions.

- *Integrated explainability framework:* We develop an integrated explainability framework that leverages the model's inherent attention mechanisms and SHAP-based feature attribution. The AttentionPooling module provides model-internal importance signals, while the fusion gating outputs (alpha values) indicate the relative contribution of each branch. This supports interpretation of detection decisions by indicating influential traffic features and branch contributions without treating these signals as causal explanations.
- *Comprehensive evaluation protocol:* We implement and validate a rigorous evaluation protocol for zero-day attack detection, including: the categorical exclusion of Shellcode and Worms from training and validation, class-weighted training, progressive learning rate scheduling (ReduceLROnPlateau), and extensive metric tracking (accuracy, F1, ROC-AUC, PR-AUC). The framework supports training on large-scale traffic data from UNSW-NB15 under a controlled category-holdout and consistent baseline evaluation protocol.

The remainder of this paper is organized as follows: Sect. 2 reviews pertinent literature on DL for intrusion detection and explainable AI (XAI). Section 3 details the ZD-HybridNet architecture and the integrated explainability framework. Section 4 presents and analyzes the qualitative analysis of the explanation outputs. Finally, Sect. 5 concludes the paper and suggests directions for future research.

2 Literature review

The rapid growth of IoT, IIoT, IoV, and cyber-physical systems has greatly increased the attack surface of today's networks, and the problem of zero-day attacks has become one of the most urgent issues in network security. Unlike signature-based attacks, zero-day attacks exploit hitherto unknown vulnerabilities, making traditional intrusion detection systems (IDSs) ineffective. Consequently, recent studies have begun to focus on improved machine learning (ML), deep learning (DL), transformer-based models, and explainable AI (XAI) methods to enhance the accuracy, flexibility, and reliability of detection. This review paper is a synthesis of more recent and high-impact journal articles on zero-day and novel attacks detection, which are grouped into consistent methodological groups to ensure that the discussion has a logical flow.

2.1 Deep learning–based intrusion detection approaches

One of the primary goals of the early research into deep learning-based IDS was to take advantage of the fact that neural networks, with their representation learning ability, can do the job of neural networks that traditional machine learning methods employed. CNNs, RNNs, and their hybrid forms have been widely studied, as they can learn hierarchical and discriminative structures in raw or preprocessed network traffic. These models have been particularly useful in achieving non-linear relationships and sophisticated attack behavior that signature-based and shallow learning methods frequently overlook. On this, [19] showed that CNN-based IDS models are much superior to classic machine

learning algorithms, as they can learn spatial associations among features of network traffic. Their method, tested on the UNSW-NB15 dataset, already showed significant improvements in precision, recall, and F1-score, demonstrating the appropriateness of CNNs for detecting more complex and previously unknown attack patterns. Nevertheless, CNNs are highly effective at extracting spatial features but do not scale to the sequential dependencies in network traffic. To address efficiency and scalability issues, [20] proposed a sequential deep neural network (DNN) with an Extra Tree Classifier-based feature selection mechanism. The model achieved high accuracy by reducing the feature space to the highly informative set of 8 features and greatly reducing computational overhead. This work has shown that dimensionality reduction combined with deep learning can not only increase inference speed but also enable IDS deployment in resource-limited settings. The hybrid architectures also improved the performance of the IDS by sharing the modelling of the spatial and temporal properties of network traffic. Khorsheed et al. [21] presented a CNNRNN CNNLSTM model, which combines CNNs with spatial feature extractions and LSTMs with temporal sequence modeling to produce greater resilience against ransomware and zero-day attacks. Their results were conclusive: temporal dependency modeling is fundamental for identifying changing and stealthy threats. On the same note, [22] augmented probabilistic data structures with deep learning to support real-time traffic analysis across a backbone network, with particular attention to the trade-off between detection accuracy and computational efficiency, especially in multiclass intrusion detection. In general, these early deep learning-based solutions laid the foundation for more sophisticated IDS systems by demonstrating the effectiveness of automated feature learning and revealing critical issues in temporal modeling, scalability, and real-time execution.

2.2 Attention, transformer, and graph-based models

Recent works have used attention and transformers to address the drawbacks of traditional DL models in capturing long-range dependencies. Krishnan et al. [23] came up with an attention fusion framework, which integrates both the short and long-term attention towards explainable zero-day detection in IoT networks, which yields an almost perfect accuracy and enhanced interpretability. Sachan and Malviya [24] also showed that transformer architectures can effectively detect zero-day threats in real time (with low-latency inference). Graph-based learning has also become popular for modeling relationships between networks. Biradar et al. and Yang et al. [25, 26] incorporated GNNs with RNNs and attention mechanisms in order to jointly learn spatial and temporal features, which is superior compared to standalone DL and traditional ML models. However, their use has been restricted to real-world zero-day settings because they typically rely on large amounts of labeled data and complete network visibility. To address label dependency, self-supervised and contrastive transformer models have been proposed more recently. Koukoulis et al. [27] demonstrated that self-supervised transformer encoders are better able to generalize across datasets to enhance inter-dataset zero-day detection results.

2.3 Zero-shot, meta-learning, and continual learning approaches

To address the underlying problem of invisible attacks, mathematicians have studied the paradigms of zero-shot and meta-learning. Mirza et al. [28] proposed ZDBERTa, a

zero-shot transformer model that is able to identify unseen IoV attacks without having labeled samples. Uddin et al. [29] suggested a hierarchical IDS of IoMT, where near-edge layers are based on one-class and meta-learning systems to learn about zero-day attacks with minimal data. Flexibility is also enhanced through continuous, online learning. Rustam et al. [30] introduced MULTI-LF, a continuous learning framework that updates detection models in real time across heterogeneous environments and minimizes human involvement to a great extent. Vladov et al. [31] generalized this concept into an adaptive online training and uncertainty modeling to counteract data drift and poisoning attacks in corporate networks.

2.4 Ensemble, hybrid, and reinforcement learning–based methods

The hybrid and ensemble intrusion detection systems (IDSs) have become viable in-depth countermeasures, offering improved detection accuracy, maturity, and computational effectiveness, especially for zero-day attack detection. By combining several learning models, ensemble-based techniques reduce the bias and variance of individual classifiers and improve generalization to new types of attacks. Zaki and Naser [32] suggested a hybrid ensemble IDS, which incorporates XGBoost, KNN, and SGD classifiers by using their joint capabilities to identify zero-day DDoS attacks in IoT networks. The framework used feature selection and parallel learning to achieve such high detection accuracy with low latency, making it suitable for near-real-time applications. However, these monitored ensemble models may fail to fully defend against completely new attack patterns and remain highly sensitive to dataset fine-tuning. In an attempt to enhance flexibility in dynamic networks, Korba et al. [33] proposed a meta-ensemble construction that relies on several Isolation Forest models that are utilized in the network edge when dealing with Internet of Vehicles (IoV) networks. The stacking strategy based on PSO allowed to improve resilience against known attacks and allow decentralizing the process of detection by aggregating anomaly scores. However, the framework showed relatively poor performance against zero-day attacks, highlighting the underlying shortcomings of unmonitored anomaly detection in the face of highly complex traffic patterns. In addition to ensemble learning, reinforcement learning (RL) has attracted attention for enabling adaptive, self-learning IDSs. The DQ-IDS developed by [34] uses Deep Q-Networks to learn optimal detection policies dynamically via network feedback. The model produced lower false positives and false negatives than standard ML-based IDSs by continually updating its decision strategy, and this finding underscores the potential of RL-driven IDSs in evolving threat environments.

2.5 Explainable AI and trustworthy IDS frameworks

As intrusion detection systems are increasingly deployed in safety-critical and regulated environments, such as industrial control systems, healthcare IoT, and smart infrastructure, explainability has emerged as a fundamental requirement alongside detection accuracy. Explainable AI (XAI) techniques enable security analysts to understand, validate, and trust model decisions, particularly when responding to zero-day threats. Alabdulatif [35] proposed an explainable ensemble-based IDS that integrates ANN and SVM as base learners with a Random Forest meta-classifier, leveraging SHAP to provide both global and local feature importance, thereby enhancing decision transparency without compromising real-time performance. In contrast, [36] introduced an unsupervised and

self-learning framework that relies solely on normal traffic, employing adaptive thresholding and SHAP explanations to identify zero-day anomalies while minimizing labeling requirements. Furthermore, attention-driven explainable architectures have shown strong promise for real-time deployment. Sowmya and Konduri, Hleha and Hol [37, 38] demonstrated that attention mechanisms not only highlight critical temporal and spatial features but also maintain low inference latency, enabling accurate, interpretable, and operationally feasible intrusion detection in dynamic network environments.

2.6 Federated, edge, and large-scale deployment-oriented frameworks

The need to scale and address privacy has been a major factor in the adoption of federated and edge-based intrusion detection system (IDS) architectures, especially in large-scale Industrial IoT (IIoT) and heterogeneous enterprise-IoT applications. Older centralized IDS systems often perform poorly due to high communication overhead, latency bottlenecks, and privacy concerns about aggregating sensitive network traffic at a single point. To overcome the constraints, [39] introduced a federated IDS architecture based on a lightweight LightGBM, which is only available in the IIoT environment. The method they use enables training the model collaboratively across distributed edge nodes without sharing raw data, preserving privacy while maintaining real-time detection. The framework also indicates that federated learning can achieve ultra-low detection latency and high accuracy, making it applicable in time-sensitive industrial settings where speed and privacy are crucial.

[40], on the other hand, focused on scalability and operational deployment, designing a scalable network security traffic analysis platform applicable to enterprise and IoT networks. Their framework combines machine learning, deep learning, and natural language processing to support real-time monitoring and offline analysis. The platform supports the use of a variety of analytical tools within a single architecture, emphasizing the practical aspects of system integration, data processing pipelines, and performance across a range of datasets. All these studies highlight that scalable, privacy-conscious IDS systems must be accurate in detection, computationally efficient, and realistic for deployment (Table 1).

In general, the literature demonstrates a transition from traditional DL-based IDSs to transformer-based, explainable, and adaptive designs for zero-day attack detection. Although recent models are quite accurate, several critical issues remain, including the ability to generalize to entirely unknown attacks, computational cost, data dependence, and the complexity of deployment in the real world. The existence of these gaps underscores the need to pursue further research on lightweight, explainable, and continuously learning IDS architectures that perform well in dynamic, resource-constrained settings.

3 Methodology

This paper introduces a zero-day-aware, model-driven deep learning methodology for network intrusion detection that rigorously tests model generalization under previously unseen attack conditions. In contrast to conventional designs that rely on fixed train–test splits, our methodology explicitly defines zero-day threat modeling and leakage-free data partitioning to better reflect real-world deployment. The end-to-end workflow is summarized in Fig. 1, including data integrity checks, zero-day labeling, leakage-free partitioning, feature preprocessing, unsupervised anomaly augmentation, hybrid model

Table 1 Summary of recent Q1 studies on zero-day and advanced intrusion detection systems

Year	Refs.	Model/Framework	Key results	Limitations
2025	[23]	Attention Fusion IDS	99.99% binary accuracy with improved interpretability	Limited zero-day performance; dataset dependency
2025	[25]	GNN–RNN–Attention IDS	97.5% accuracy using spatial–temporal learning	Requires large labeled datasets
2025	[41]	ZeroDay-LLM	97.8% accuracy with reduced false positives	High transformer overhead
2025	[28]	ZDBERTa	99.31% accuracy with GAN augmentation	Synthetic data dependency
2025	[29]	Hierarchical IoMT IDS	99.77% accuracy for near-edge detection	Deployment complexity
2025	[30]	MULTI-LF	0.999 accuracy with continuous learning	High computational cost
2026	[21]	CNN–LSTM Hybrid IDS	97.89% accuracy for ransomware detection	SMOTE dependency
2025	[20]	Sequential DNN IDS	97.93% accuracy with reduced features	Weak generalization to novel attacks
2024	[32]	Hybrid Ensemble IDS	99.91% accuracy with low latency	Dataset-specific tuning
2024	[42]	MLP + XAI IDS	99.62% accuracy on KDD99	Outdated dataset
2025	[39]	Federated LightGBM IDS	99.8% accuracy with low latency	Federated overhead
2026	[36]	ADAPT-IDS	Improved zero-day detection	Moderate recall
2024	[43]	Hybrid Zero-Day IDS	98.4% online accuracy	Threshold sensitivity
2025	[35]	Explainable Ensemble IDS	99.40% accuracy with SHAP	Dataset dependency
2025	[31]	Adaptive IDS/IPS	ROC-AUC of 0.96 with drift robustness	Architectural complexity

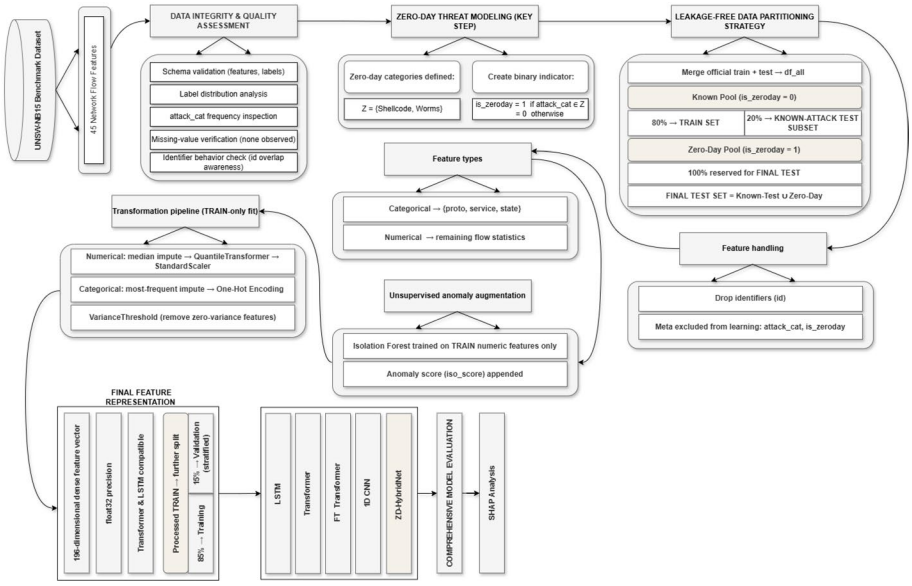


Fig. 1 End-to-end zero-day–aware intrusion detection workflow on UNSW-NB15, illustrating integrity checks, zero-day threat definition, leakage-free partitioning, preprocessing with anomaly augmentation, hybrid model learning, and evaluation

training, and evaluation. We ensure that every stage of the pipeline prevents exposure of designated zero-day categories during training, enabling fair assessment of detection performance on truly unseen attack types (Fig. 1). Each component is detailed in the following subsections and summarized in Fig. 1.

3.1 Dataset description

This study employs the UNSW-NB15 dataset [44], a widely used benchmark for network intrusion detection research. The dataset was generated using the IXIA PerfectStorm tool in the Cyber Range Lab of the Australian Centre for Cyber Security (ACCS) at UNSW Canberra to simulate realistic, benign network activities alongside contemporary attack behaviors. Approximately 100 GB of raw network traffic was recorded using the Tcpdump tool and subsequently processed with Argus and Bro-IDS to extract flow-level features. The dataset includes nine attack categories, namely Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms, in addition to normal traffic. For each network flow, 49 statistical and protocol-level features are derived and documented in the official UNSW-NB15 feature specification. In total, the dataset contains 2,540,044 records and provides predefined training and testing splits consisting of 82,332–175,341 samples, respectively. In this work, the official splits are used only as an initial reference. They are subsequently restructured to support a strict zero-day evaluation protocol, where selected attack categories are entirely excluded from the training phase and reserved exclusively for final evaluation. This design enables a realistic assessment of model generalization under unseen attack conditions, consistent with the workflow in Fig. 1.

3.1.1 Data integrity and quality assessment

We first assess the structural integrity and quality of the UNSW-NB15 dataset. The existence and consistency of the binary class label (`label`) and the attack category attribute (`attack_cat`) are verified across all samples. The empirical class distribution is examined as:

$$P(y = c) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(y_i = c), \quad c \in \{0, 1\}, \tag{1}$$

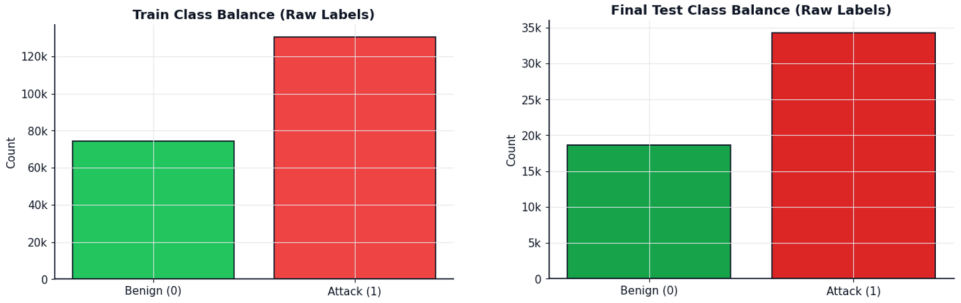
where $y = 0$ denotes benign traffic and $y = 1$ denotes attack traffic. In addition, the frequency of individual attack categories is analyzed to identify potential imbalance:

$$\text{Count}(k) = \sum_{i=1}^{N} \mathbb{I}(\text{attack_cat}_i = k). \tag{2}$$

A column-wise inspection confirms the absence of missing values across all features. Furthermore, the identifier attribute (`id`) is observed to overlap between the official training and testing partitions, indicating that it functions as an index rather than a discriminative traffic feature. To prevent unintended information leakage, this attribute is removed prior to model training. The resulting class balance after restructuring is shown in Fig. 2.

3.1.2 Zero-day threat modeling

To evaluate generalization under previously unseen attack conditions, this study adopts a controlled category-holdout zero-day setting. The low-frequency attack categories Shellcode and Worms were selected because their limited representation in the UNSW-NB15 dataset provides a challenging basis for evaluating whether malicious traffic from categories that are completely absent during model development can still be detected.



(a) Train set class balance (raw labels). (b) Final test set class balance (raw labels).

Fig. 2 Class distribution in training and final test sets after restructuring under the proposed zero-day evaluation protocol

Both categories are excluded from the training and validation subsets and are retained exclusively for final testing.

The zero-day attack set is formally defined as:

$$\mathcal{Z} = \{\text{Shellcode}, \text{Worms}\}. \tag{3}$$

A binary indicator variable is constructed for each network flow:

$$\text{is_zeroday}_i = \begin{cases} 1, & \text{if attack_cat}_i \in \mathcal{Z}, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

This indicator is used exclusively for dataset partitioning and evaluation purposes and is never exposed to the learning model.

Accordingly, the term "zero-day" in this study refers to attack categories that are unseen during training rather than to newly discovered software vulnerabilities. The proposed model performs binary benign-versus-attack detection; therefore, it evaluates whether samples belonging to the held-out categories can be identified as malicious, but it does not recognize or assign their specific attack-family labels.

The present evaluation is focused on the defined Shellcode–Worms holdout configuration, which provides a controlled and leakage-free setting for assessing unseen-category attack detection. Evaluation under additional category-holdout configurations is beyond the scope of the current experimental design.

The proportion of zero-day samples present in the final test set is visualized in Fig. 3.

3.1.3 Leakage-controlled data partitioning

To avoid category leakage between training and evaluation, the official UNSW-NB15 training and testing partitions are first merged into a unified dataset:

$$\mathcal{D}_{all} = \mathcal{D}_{train}^{official} \cup \mathcal{D}_{test}^{official}. \tag{5}$$

Following the merge, exact row-wise duplicate screening was performed before any partition was generated, preventing identical flow records from being distributed across the development and evaluation subsets. All subsequent partitions were created at the individual flow-record level. Accordingly, the independence assumption adopted in this study applies to individual network-flow records rather than to session-grouped or temporally grouped traffic.

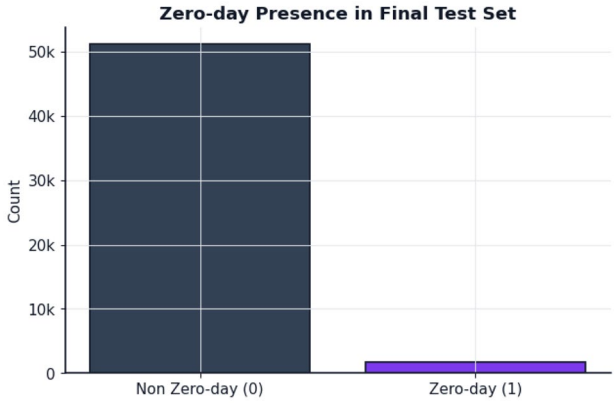


Fig. 3 Distribution of category-held-out zero-day samples in the final test set. Shellcode and Worms are completely excluded from model development and included only during final evaluation, illustrating their limited proportion relative to the remaining test samples

Table 2 Data partitioning strategy used for zero-day intrusion detection after merging the official splits and applying exact duplicate screening

Subset	Description	Samples	Purpose
Official Train Set	Provided UNSW-NB15 training split	82,332	Initial data source
Official Test Set	Provided UNSW-NB15 testing split	175,341	Initial data source
Merged Dataset ($\mathcal{D}_{all}$)	Train ∪ Test (official)	257,673	Unified pool
Known-Attack Pool ($\mathcal{D}_{known}$)	All non-zero-day samples	255,988	Model development
Zero-Day Pool ($\mathcal{D}_{Z}$)	Shellcode + Worms attacks	1685	Unseen attack evaluation
Training Pool ($\mathcal{D}_{train}$)	80% of $\mathcal{D}_{known}$ (stratified)	204,790	Training and validation partitioning
Final Training Subset ($\mathcal{D}_{train}^{85}$)	85% of the training pool	174,071	Model training
Validation Set ($\mathcal{D}_{val}^{15}$)	15% of the training pool	30,719	Hyperparameter tuning
Known-Test Subset ($\mathcal{D}_{test}^{known}$)	20% of $\mathcal{D}_{known}$	51,198	Known-attack testing
Final Test Set ($\mathcal{D}_{test}^{final}$)	Known-Test ∪ Zero-Day	52,883	Final evaluation

The full partitioning strategy and sample counts are summarized in Table 2. The known-attack pool was first divided using an 80/20 row-level stratified random split based exclusively on the binary class label, with a fixed random seed of 42. This procedure preserves the benign-to-attack class proportion while reserving a sufficiently large known-attack subset for final evaluation.

The resulting 80% subset was treated as the training pool and was subsequently divided using an 85/15 stratified random split, again based on the binary class label and the same random seed. The resulting 85% subset was used for model training, whereas the remaining 15% was used for hyperparameter selection and early stopping. Both stages were completed before the final test set was accessed.

Attack-category labels were not used as stratification variables; they were retained only to enforce and verify zero-day category isolation. Post-partition verification confirmed that Shellcode and Worms were absent from both the final training and validation subsets and were included exclusively in the final test set.

The merged dataset is then separated into a known-attack pool and a zero-day pool as follows:

$$\mathcal{D}_{known} = \{(x_i, y_i) \in \mathcal{D}_{all} \mid \text{is_zeroday}_i = 0\}, \tag{6}$$

$$\mathcal{D}_Z = \{(x_i, y_i) \in \mathcal{D}_{all} \mid \text{is_zeroday}_i = 1\}. \tag{7}$$

The known-attack pool is divided at the flow-record level using an 80/20 stratified random split based on the binary class label and a fixed random seed of 42:

$$(\mathcal{D}_{train},\ \mathcal{D}_{test}^{known}) = \text{StratifiedSplit}(\mathcal{D}_{known},\ 0.8,\ 0.2;\ y,\ 42). \tag{8}$$

The training pool is then divided into the final training and validation subsets using an 85/15 stratified split under the same reproducibility setting:

$$(\mathcal{D}_{train}^{85},\ \mathcal{D}_{val}^{15}) = \text{StratifiedSplit}(\mathcal{D}_{train},\ 0.85,\ 0.15;\ y,\ 42). \tag{9}$$

All zero-day samples are excluded from model development and appended exclusively to the final test set:

$$\mathcal{D}_{test}^{final} = \mathcal{D}_{test}^{known} \cup \mathcal{D}_Z. \tag{10}$$

To enforce strict category-level zero-day isolation, the following constraints are applied:

$$\mathcal{A}(\mathcal{D}_{train}^{85}) \cap \mathcal{Z} = \emptyset, \qquad \mathcal{A}(\mathcal{D}_{val}^{15}) \cap \mathcal{Z} = \emptyset, \tag{11}$$

where $\mathcal{A}(\cdot)$ denotes the set of attack categories present in a dataset.

These constraints ensure that the designated zero-day categories are not observed during model training, hyperparameter selection, or early stopping. They are introduced only during final evaluation through $\mathcal{D}_{test}^{final}$.

As reported in Table 3, all known attack categories are represented across the final training, validation, and known-test subsets under binary-label stratification. In contrast, Shellcode and Worms contain no samples in the final training, validation, or known-test subsets and are retained exclusively in the zero-day test subset. The reported distribution therefore confirms category-level isolation of the designated zero-day attacks while preserving the overall benign-to-attack proportions used during model development and evaluation.

The resulting training versus final test class distributions are shown in Fig. 2, while the zero-day proportion in the final test set is shown in Fig. 3.

Table 3 Attack-category distribution across the final training, validation, known-test, and zero-day test subsets

Attack category	Final train	Validation	Known test	Zero-day test
Normal	63,240	11,160	18,600	0
Fuzzers	16,525	2888	4833	0
Analysis	1826	342	509	0
Backdoor	1574	261	494	0
DoS	11,213	1912	3228	0
Exploits	30,179	5412	8934	0
Generic	40,062	7074	11,735	0
Reconnaissance	9452	1670	2865	0
Shellcode	0	0	0	1,511
Worms	0	0	0	174
Total	**174,071**	**30,719**	**51,198**	**1685**

Bold values indicate column totals

3.1.4 Feature preprocessing and representation learning

Before feature transformation, non-informative and meta-level attributes (id, attack_cat, is_zeroday) are excluded from the learning input, consistent with the category-isolation constraint in Eq. 11. The remaining attributes are categorized into categorical features (proto, service, state) and numerical flow-level statistics.

All preprocessing components were fitted exclusively on the final training subset $\mathcal{D}_{train}^{85}$. The fitted transformations were subsequently applied without re-estimation to the validation, known-test, and zero-day test subsets, thereby preventing evaluation-set information from influencing feature construction.

To provide an unsupervised descriptor of distributional deviation, an Isolation Forest is used to generate an additional anomaly-score feature. Although class labels are available, the anomaly score is learned without supervision to capture distributional irregularities that may not align with known attack classes:

$$s_i = f_{\mathrm{IF}}(x_i^{num}), \tag{12}$$

where $f_{\mathrm{IF}}(\cdot)$ denotes the Isolation Forest decision function.

The Isolation Forest was configured with 100 estimators, `max_samples=auto`, `contamination=auto`, and a fixed random seed of 42. It was fitted only on the numerical attributes of $\mathcal{D}_{train}^{85}$, and the resulting decision score was appended as an additional numerical feature for all data subsets.

Numerical features undergo median imputation followed by quantile transformation:

$$\tilde{x}_{ij} = \Phi^{-1}\big(F_j(x_{ij})\big), \tag{13}$$

where $F_j(\cdot)$ represents the empirical cumulative distribution function estimated on the final training subset and Φ^{-1} denotes the inverse standard normal distribution. This transformation mitigates extreme skewness commonly observed in network traffic and improves numerical stability for gradient-based learning.

The quantile transformation was implemented using 1000 quantiles, a normal output distribution, a subsample size of 10,000, and a fixed random seed of 42. Standard scaling was subsequently fitted on the transformed training features and applied unchanged to the remaining subsets.

Categorical features are imputed using the most frequent category and encoded via one-hot encoding.

The encoder was configured with handle_unknown=ignore, allowing previously unobserved categorical values in the validation or test subsets to be processed without modifying the training-derived feature space. Features with zero variance were then removed using a variance-threshold value of 0.0.

The complete preprocessing order was therefore: metadata removal, numerical median imputation, Isolation Forest anomaly-score generation, quantile transformation, standard scaling, categorical mode imputation, one-hot encoding, and zero-variance feature removal. The same fitted preprocessing pipeline and resulting feature representation were used for ZD-HybridNet and all baseline models to ensure a consistent experimental comparison.

The transformed feature order, original numerical feature names, generated one-hot encoded category names, and the appended Isolation Forest score were retained in a deterministic feature-index mapping. This mapping enables the anonymous processed

feature indices used during model computation to be traced back to their corresponding UNSW-NB15 attributes in the subsequent SHAP analysis.

The effect of preprocessing on two representative skewed features is illustrated in Fig. 4.

3.1.5 Final feature representation

After preprocessing, each network flow is represented as a dense vector of 196 features with `float32` precision.

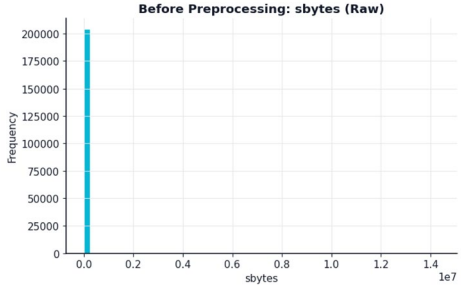
The final training and validation subsets follow the stratified 85/15 partitioning procedure defined in Eq. 9; therefore, the split is not repeated at this stage.

The resulting 196-dimensional representation preserves a fixed feature order across all data subsets and is used consistently as the input to ZD-HybridNet and all baseline architectures. This ensures that the comparative models are evaluated using an identical feature space and preprocessing configuration.

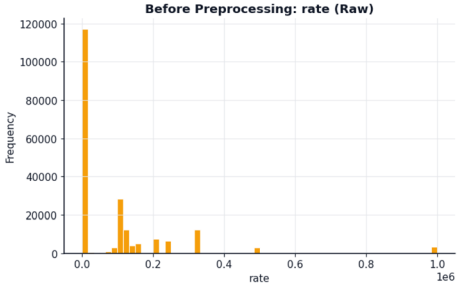
Given the class imbalance observed in Fig. 2, inverse-frequency class weights are computed exclusively from the final training subset $\mathcal{D}^{85}_{train}$ and are applied only during model optimization. Validation, known-test, and zero-day test labels are not used to estimate these weights. The resulting class weights are shown in Fig. 5.

3.2 Proposed ZD-HybridNet architecture

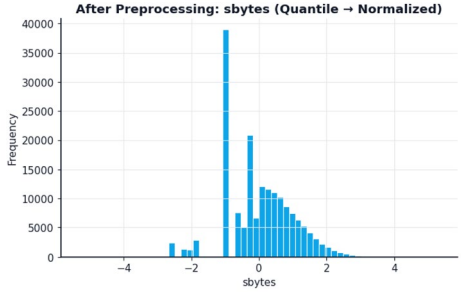
This work proposes **ZD-HybridNet (Zero-Day Hybrid Attention Network)**, a hybrid deep learning architecture designed to address the inherent generalization challenges of zero-day intrusion detection. Unlike conventional single-branch models, ZD-HybridNet explicitly integrates global contextual modeling with sequential dependency learning



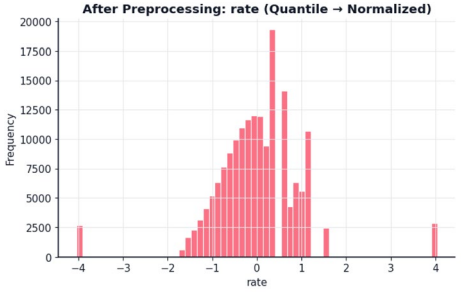
(a) `sbytes` before preprocessing (raw).



(b) `rate` before preprocessing (raw).



(c) `sbytes` after preprocessing (quantile transformed and normalized).



(d) `rate` after preprocessing (quantile transformed and normalized).

Fig. 4 Effect of preprocessing on highly skewed network traffic features. Quantile transformation (Eq. 13) reduces extreme skewness and stabilizes the feature distributions used for neural network training

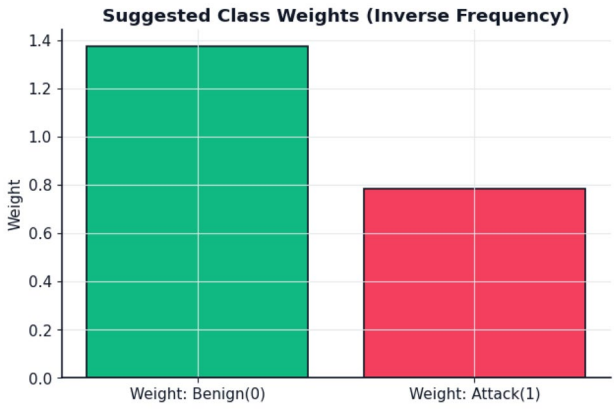


Fig. 5 Inverse-frequency class weights computed exclusively from the final training subset $\mathcal{D}_{train}^{85}$ to balance class contributions during model optimization.

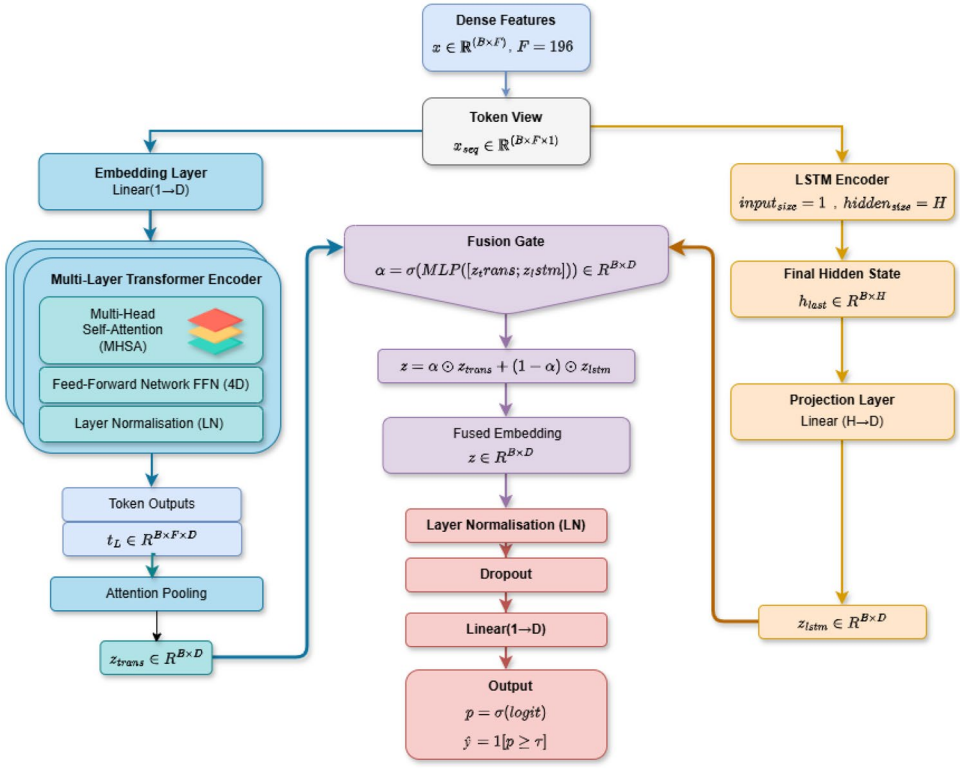


Fig. 6 Architecture of ZD-HybridNet: a zero-day-aware hybrid attention network for intrusion detection. Dense flow-level features are processed in parallel by a Transformer encoder and an LSTM encoder. An adaptive attention-based fusion mechanism integrates global contextual dependencies and sequential feature patterns before final binary classification

to capture diverse behavioral characteristics of network traffic under unseen attack conditions.

An overview of the proposed ZD-HybridNet architecture is shown in Fig. 6. The model operates on dense, preprocessed network flow representations and produces a probabilistic binary decision indicating malicious activity.

3.2.1 Input representation and tokenization

Each network flow is represented by a dense feature vector

$$x \in \mathbb{R}^{B\times F}, \tag{14}$$

where B denotes the batch size and $F = 196$ represents the number of preprocessed features. To enable sequence-aware modeling, the feature vector is reshaped into a tokenized sequence:

$$x_{\text{seq}} \in \mathbb{R}^{B\times F\times 1}. \tag{15}$$

This formulation allows individual features to be interpreted as ordered tokens, enabling both attention-based and recurrent architectures to model inter-feature dependencies.

3.2.2 Transformer encoder branch

In the Transformer branch, each token is first projected into a D-dimensional embedding space using a linear embedding layer:

$$e = W_e x_{\text{seq}}, \quad e \in \mathbb{R}^{B\times F\times D}. \tag{16}$$

The embedded tokens are processed by a multi-layer Transformer encoder composed of multi-head self-attention, feed-forward networks, residual connections, and layer normalization. This branch captures global feature interactions and long-range dependencies across network flow attributes.

Let $\mathrm{H}_{\text{trans}} = [\mathrm{h}_1, \mathrm{h}_2, \ldots, \mathrm{h}_F] \in \mathbb{R}^{B\times F\times D}$ denote the token-level representations generated by the Transformer encoder. A learnable scoring function $g_\phi(\cdot)$ assigns an attention score to each feature token:

$$u_i = g_\phi(\mathrm{h}_i), \qquad i = 1, \ldots, F. \tag{17}$$

The scores are normalized across the feature-token dimension using the softmax function:

$$a_i = \frac{\exp(u_i)}{\sum_{j=1}^{F} \exp(u_j)}, \qquad \sum_{i=1}^{F} a_i = 1. \tag{18}$$

The Transformer representation is then obtained as the attention-weighted sum of the token-level embeddings:

$$z_{\text{trans}} = \sum_{i=1}^{F} a_i \mathrm{h}_i, \qquad z_{\text{trans}} \in \mathbb{R}^{B\times D}. \tag{19}$$

The normalized weights a_i provide sample-specific information about which transformed network-traffic features receive greater emphasis during representation aggregation. Because the processed feature indices are retained through the feature-name mapping described in 3.1.4, these weights can be traced back to the corresponding UNSW-NB15 attributes.

The attention weights are interpreted as model-internal allocation signals rather than causal explanations. They are therefore used as complementary evidence alongside the SHAP-based feature analysis presented in the results section.

3.2.3 LSTM encoder branch

In parallel, the tokenized feature sequence is processed by an LSTM encoder with input dimension 1 and hidden size H. The LSTM branch models ordered dependency patterns and localized sequential relationships among flow features:

$$h_t = \text{LSTM}(x_{\text{seq}}), \tag{20}$$

where h_t denotes the hidden state at time step t.

The final hidden state is extracted as a compact sequence representation:

$$h_{\text{last}} \in \mathbb{R}^{B\times H}. \tag{21}$$

To enable fusion with the Transformer output, a linear projection maps this representation into the shared embedding space:

$$z_{\text{lstm}} = W_p h_{\text{last}}, \quad z_{\text{lstm}} \in \mathbb{R}^{B\times D}. \tag{22}$$

3.2.4 Adaptive attention-based fusion

To combine the complementary representations learned by both branches, ZD-Hybrid-Net employs an adaptive attention-based fusion mechanism. A learnable fusion gate is computed as:

$$\alpha = \sigma\left(\text{MLP}\left([z_{\text{trans}};\, z_{\text{lstm}}]\right)\right), \tag{23}$$

where $\sigma(\cdot)$ denotes the sigmoid function, $[\cdot;\cdot]$ indicates vector concatenation, and $\alpha \in \mathbb{R}^{B\times D}$.

The gate is computed independently for each input sample and embedding dimension, with $\alpha \in [0,1]^{B\times D}$. Values approaching 1 assign greater weight to the Transformer representation, whereas values approaching 0 emphasize the LSTM representation.

The final fused representation is obtained via element-wise weighting:

$$z = \alpha \odot z_{\text{trans}} + (1-\alpha) \odot z_{\text{lstm}}, \tag{24}$$

where $\odot$ denotes Hadamard multiplication.

This sample-dependent mechanism enables the model to adaptively balance global feature interactions and sequential dependencies for each network flow. The resulting fusion coefficients also provide an internal diagnostic signal for examining the relative contribution of the two branches during inference.

3.2.5 Classification head

The fused embedding is passed through layer normalization and dropout for regularization, followed by a linear classification layer:

$$\text{logit} = W_o z + b. \tag{25}$$

The final intrusion probability is obtained using the sigmoid activation:

$$p = \sigma(\text{logit}), \tag{26}$$

and the predicted class label is defined as:

$$\hat{y} = \begin{cases} 1, & p \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{27}$$

where τ is a decision threshold selected based on validation performance.

Zero-day attacks often manifest as subtle deviations that are not well captured by purely local or purely global models. The Transformer branch of ZD-HybridNet effectively models global feature interactions, while the LSTM branch captures ordered dependency patterns across flow attributes. The adaptive fusion mechanism enables the model to dynamically emphasize the most informative representation, resulting in improved robustness and generalization under unseen attack scenarios.

Require: Official UNSW-NB15 splits $\mathcal{D}_{train}^{official}, \mathcal{D}_{test}^{official}$; zero-day set $\mathcal{Z} = \{\text{Shellcode}, \text{Worms}\}$
Ensure: Trained model θ^*; decision threshold τ^*; metrics on $\mathcal{D}_{test}^{final}$
Merge and annotate:
$\mathcal{D}_{all} \leftarrow \mathcal{D}_{train}^{official} \cup \mathcal{D}_{test}^{official}$; remove exact duplicate rows
for each sample (x_i, y_i) in $\mathcal{D}_{all}$ **do**
  $\text{is_zeroday}_i \leftarrow \mathbb{I}(\text{attack_cat}_i \in \mathcal{Z})$
end for
Leakage-controlled partitioning:
$\mathcal{D}_{known} \leftarrow \{(x_i, y_i) \in \mathcal{D}_{all} : \text{is_zeroday}_i = 0\}$
$\mathcal{D}_Z \leftarrow \{(x_i, y_i) \in \mathcal{D}_{all} : \text{is_zeroday}_i = 1\}$
$(\mathcal{D}_{train}, \mathcal{D}_{test}^{known}) \leftarrow \textsc{StratifiedSplit}(\mathcal{D}_{known}, 0.8, 0.2; 42)$
$(\mathcal{D}_{train}^{85}, \mathcal{D}_{val}^{15}) \leftarrow \textsc{StratifiedSplit}(\mathcal{D}_{train}, 0.85, 0.15; 42)$
$\mathcal{D}_{test}^{final} \leftarrow \mathcal{D}_{test}^{known} \cup \mathcal{D}_Z$
Enforce $\mathcal{A}(\mathcal{D}_{train}^{85}) \cap \mathcal{Z} = \varnothing$ and $\mathcal{A}(\mathcal{D}_{val}^{15}) \cap \mathcal{Z} = \varnothing$
Preprocessing (fit on final train only):
Drop meta columns: `id`, `attack_cat`, `is_zeroday`
Fit Isolation Forest on numeric features of $\mathcal{D}_{train}^{85}$; append anomaly score s_i
Fit transforms on $\mathcal{D}_{train}^{85}$:
  Numeric: median impute $\rightarrow$ QuantileTransform $\rightarrow$ StandardScaler
  Categorical: mode impute $\rightarrow$ OneHot; then VarianceThreshold
Transform $\mathcal{D}_{train}^{85}$, $\mathcal{D}_{val}^{15}$, and $\mathcal{D}_{test}^{final}$ to dense features $x \in \mathbb{R}^{196}$
Train proposed hybrid model:
Initialize parameters θ using random seed 42
for *epoch* $= 1$ to E **do**
  Update θ on mini-batches from $\mathcal{D}_{train}^{85}$ using AdamW (weighted BCE)
  Evaluate validation loss on $\mathcal{D}_{val}^{15}$; apply early stopping
end for
$\theta^* \leftarrow$ best checkpoint by validation criterion
Threshold selection (validation):
Compute validation probabilities $p_i \leftarrow \sigma(f_{\theta^*}(x_i))$ for $(x_i, y_i) \in \mathcal{D}_{val}^{15}$
$\tau^* \leftarrow \arg\max_{\tau \in [0,1]} \text{F1}\big(y, \mathbb{I}(p \geq \tau)\big)$
Final evaluation:
Compute metrics on $\mathcal{D}_{train}^{85}$, $\mathcal{D}_{val}^{15}$, and $\mathcal{D}_{test}^{final}$ using τ^*
return θ^*, τ^*, and test metrics

Algorithm 1 Zero-day–aware training and evaluation of the proposed Hybrid IDS

4 Result and discussion

This section constitutes a thorough analysis of the hybrid model of cyber attack identification by focusing on a vast amount of experimental analysis. The outcomes are planned to evaluate the effectiveness, robustness, and the ability of the model to be generalized to many evaluation perspectives, such as the overall performance of the detection, attack-wise behavior, and comparative analysis with the methods of the basis.Confusion

matrices, precision-recall curves, ROC curves, and feature importance interpretations have been examples of qualitative visual reports. Standard classification metrics, comprising accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), are used to provide quantitative performance reports. All these findings can be used to discover the predictive power and reliability of the proposed framework in making decisions.

In order to achieve a fair and reproducible test, every experiment was run within the same set of preprocessing and experimental conditions on training, validation and test sets. Comparison outcomes show that the given model is better than the current practices, and the strengths and error estimates prove that it can be applied in real-world conditions of cyber attack detection. Altogether, the given results confirm the efficiency and the feasibility of the offered strategy.

4.1 Experimental setup and evaluation metrics

The proposed hybrid framework was implemented and evaluated in a cloud-based experimental environment using Google Colab Pro with an NVIDIA Tesla T4 GPU. The implementation was developed in Python 3.10 using *TensorFlow/Keras* for model construction and training, *NumPy* and *Pandas* for numerical computation and data handling, and *Scikit-learn* for preprocessing and performance evaluation. *SHAP* was used for feature-level explanation analysis. A fixed random seed of 42 was applied to data partitioning, preprocessing components with stochastic operations, and model initialization to improve experimental reproducibility. The same computing environment, preprocessing pipeline, data partitions, class-weighting strategy, and validation-based threshold-selection procedure were used for ZD-HybridNet and all baseline models. This environment supported the controlled training and evaluation of the binary intrusion-detection architecture on the large-scale UNSW-NB15 network traffic dataset.

4.2 Zero-day-specific detection performance

To examine whether the aggregate test performance reflects effective detection of the held-out attack categories, the final test predictions were analyzed separately for known attacks, all zero-day attacks, Shellcode, and Worms. Each attack group was evaluated against the same benign test subset, ensuring a consistent binary benign-versus-attack comparison.

As summarized in Table 5, The subgroup analysis distinguishes the model's performance on attack categories observed during training from its performance on the held-out Shellcode and Worms categories. The known-attack subset achieves a recall of 0.9426, whereas the combined zero-day subset retains a recall of 0.8813. Shellcode exhibits stronger detection than Worms, indicating that the rarer Worms category presents a more difficult unseen-category detection scenario. These findings provide a more direct assessment of zero-day generalization than the aggregate final-test metrics alone.

4.3 Comparative evaluation using performance metrics

A controlled comparative evaluation was conducted using LSTM, Transformer, FT-Transformer, and 1D CNN baseline models. All models were evaluated using the same data partitions, 196-dimensional feature representation, preprocessing pipeline, class-weighting strategy, early-stopping criterion, and validation-based threshold-selection

Table 4 Final hyperparameter configuration of the proposed ZD-HybridNet model

Component	Hyperparameter setting
Input	Dense feature dimension $F = 196$ (`float32`)
Tokenization	Token length $= F$, token channel $= 1$
Transformer encoder	Layers $L = 2$; model dimension $D = 128$; heads $h = 4$
	FFN hidden dimension $= 4D = 512$; activation = GELU
	Dropout $= 0.20$; LayerNorm = pre-norm
Pooling	Attention pooling to obtain $z_{\text{trans}} \in \mathbb{R}^{B\times D}$
LSTM encoder	Layers $= 1$; hidden size $H = 128$; dropout $= 0.20$
Projection	Linear $(H \to D)$ to obtain $z_{\text{lstm}} \in \mathbb{R}^{B\times D}$
Fusion gate	MLP: input 2*D*, hidden *D*, output *D*; sigmoid gate α
Classifier head	LayerNorm + Dropout(0.20) + Linear $(D \to 1)$
Optimization	AdamW; learning rate 1×10^{-3}; weight decay 1×10^{-4}
Training	Final selected batch size $= 90$; max epochs $E = 50$; early stopping patience $= 7$
Loss	Weighted binary cross-entropy using class weights computed from $\mathcal{D}_{train}^{85}$
Threshold	Validation-selected threshold $\tau^* = 0.35$, determined using $\mathcal{D}_{val}^{15}$
Reproducibility	Fixed random seed $= 42$ for data partitioning and model training

Table 5 Detection performance for known and category-held-out zero-day attack groups

Evaluation group	Attack samples	TP	FN	TN	FP	Precision	Recall
Known attacks	32,598	30,726	1872	17,522	1078	0.9661	0.9426
All zero-day attacks	1685	1485	200	17,522	1078	0.5794	0.8813
Shellcode	1511	1360	151	17,522	1078	0.5578	0.9001
Worms	174	125	49	17,522	1078	0.1039	0.7184

Table 6 Performance comparison of the baseline models and ZD-HybridNet

Model	Accuracy	Precision	Recall	F1-score
LSTM	0.9152	0.9348	0.9109	0.9275
Transformer	0.9236	0.9429	0.9203	0.9363
FT-Transformer	0.8928	0.9185	0.8912	0.9046
1D CNN	0.8869	0.9023	0.8837	0.8978
ZD-HybridNet	**0.9404**	**0.9676**	**0.9396**	**0.9534**

Bold values indicate the best performance in each metric

procedure. The final architectural and optimization configuration of ZD-HybridNet is reported in Table 4. To ensure a controlled comparison, all baseline models received the same 196-dimensional input representation and were trained using identical data partitions, preprocessing transformations, class-weighting strategy, AdamW optimizer, learning rate of 1×10^{-3}, weight decay of 1×10^{-4}, batch size of 90, maximum of 50 epochs, early-stopping patience of 7, and random seed of 42. The decision threshold for each model was selected exclusively from the validation subset using the same F1-based criterion. Consequently, the reported differences primarily reflect the representation-learning capability of the respective architectures rather than variations in the evaluation protocol.

As reported in Table 6, ZD-HybridNet achieved the highest accuracy (0.9404), precision (0.9676), recall (0.9396), and F1-score (0.9534) among the baseline models evaluated under the same experimental protocol. The standalone Transformer provided the strongest baseline performance, followed by the LSTM, whereas the FT-Transformer and 1D CNN produced comparatively lower scores.

The observed improvement indicates that combining global feature interactions from the Transformer branch with sequential dependencies from the LSTM branch is beneficial within this controlled baseline comparison. However, these results are limited to models evaluated using the same dataset, partitioning protocol, preprocessing pipeline, and binary detection task and do not imply universal superiority over methods reported under different experimental conditions.

Figure 7 presents a direct metric-wise comparison, while Fig. 8 provides a complementary view of the balance among accuracy, precision, recall, and F1-score. Both visualizations show that ZD-HybridNet maintains the strongest and most balanced performance among the evaluated baseline architectures.

4.4 Attack-wise detection analysis

Precisionrecall (PR) curves were used to further analyze the results of the attack-wise detection capability of the proposed hybrid model with respect to training, validation, and test data to determine the performance of the classes separately under varying data splits.

Based on the results in Fig. 9a, the PR curve in the training set has an almost-perfect performance of both attack and benign classes with an average precision of 0.9955–0.9858, respectively. This means that discriminative patterns have been learned effectively in an unbiased way and not on a particular class. The validation of the results in Fig. 9b shows the same tendency with high AP values of 0.9945–0.9828 with attack and benign traffic respectively, indicating good generalization properties and little overfitting.

The PR curve on the test set that was plotted in Fig. 9c is another indication of the strength of the proposed model when the data is not visible. The attack type scores 0.9942, whereas the benign type scores 0.9814, and these indicate that the performance of the model in identifying attacks is reliable at high levels of recall. This minor loss of precision at the end extremes of recall is reasonable in the context of real-life cybersecurity and is not high.

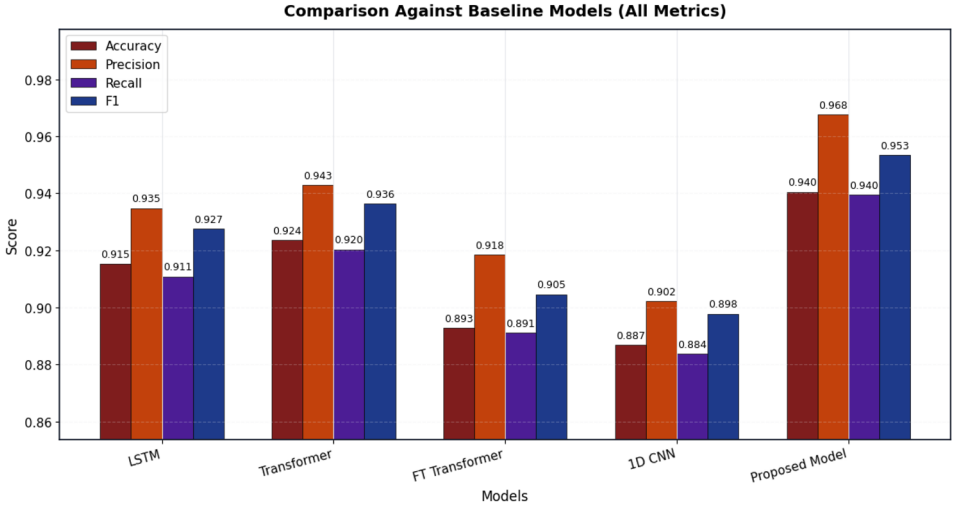


Fig. 7 Performance comparison of ZD-HybridNet and the baseline models

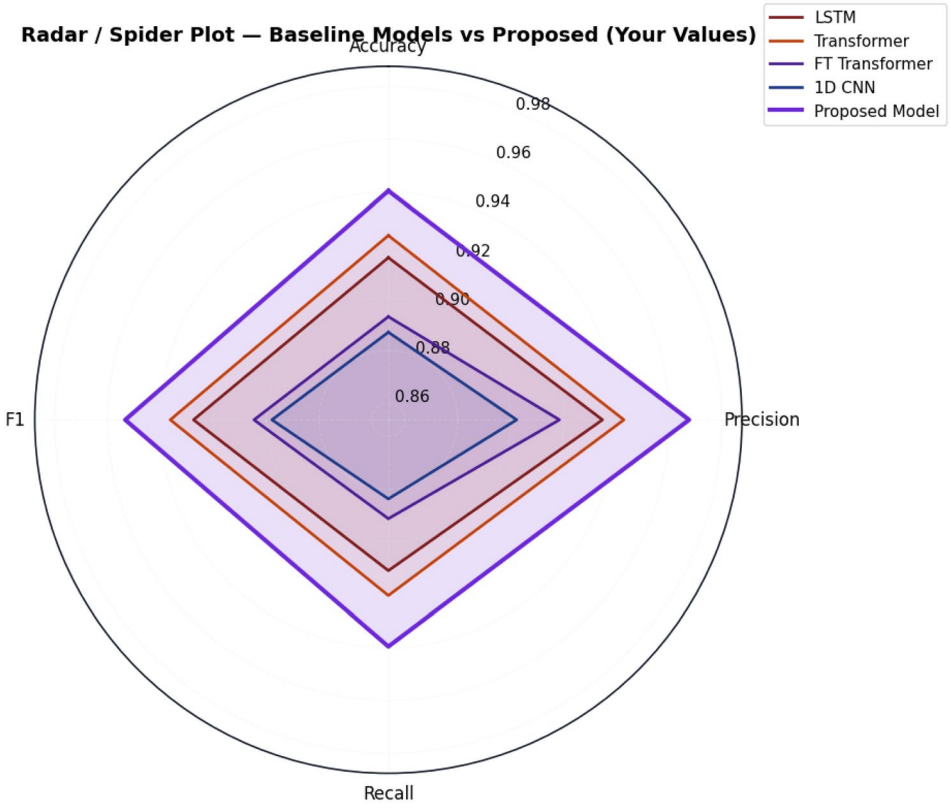
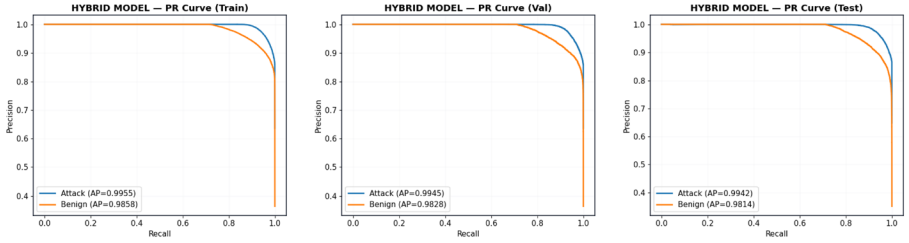


Fig. 8 Radar-based performance comparison of ZD-HybridNet and the baseline models



(a) Training PR Curve (b) Validation PR Curve (c) Testing PR Curve

Fig. 9 Precision-Recall analysis across data splits, highlighting the Average Precision (AP) for Attack and Benign classes

All in all, the high PR performance measured throughout the entire data splits indicates that the suggested model is useful in distinguishing accurately between attack and benign traffic, which is why it is applicable in the real-world cyber attack detection systems.

4.5 Confusion matrix and error analysis

The confusion matrices were used to have further information on the classification behaviour and error behaviour of the proposed hybrid model on a training, validation and test set of data. This analysis has reflected the capability of the model to appropriately differentiate benign and attack traffic and the possible sources of misclassification.

Figure 10a depicts the confusion matrix of the training data, and it shows that there is a high learning potential with a lot of benign (60,138) and attack samples (104,644) being correctly classified. The comparatively low values of false positives (3 102 benign samples identified as attack) and false negatives (6 187 attack samples identified as benign) suggest that the classes are effectively discriminated, and the laws of minimal bias are introduced in the process of training.

The validation confusion matrix presented in Fig. 10b proves the generalization capability of the model. The proportion of correctly identified benign (10,504) and attack (18,405) instances is high and a limited number of misclassifications. The lower values of false positive (656) and false negative (1154) in terms of the training data indicates the stable performance and controlled overfitting.

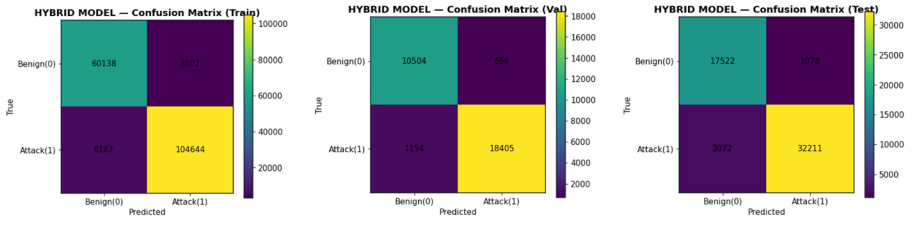
The test confusion matrix implied in Fig. 10c also supports even stronger the robustness of the proposed model in the situation of unseen data. This model is able to recognize 32,211 samples of attack in the correct amount, 17,522 samples of benign in the correct amount with relatively low false positive (1078) and false negative (2072) rates. It is important to note that false negatives, which are essential in a cybersecurity-related application, are limited, which proves the effectiveness of the model in reducing the attack detection failures.

Altogether, the data of a confusion matrix analysis of all the data splits prove that the proposed hybrid model finds the high detection accuracy with the equal error distribution. The low misclassification rates and the lack of variation in the results of training, validation, and testing steps indicate the high level of reliability and usability in the real world situation of cyber attack detection.

4.6 ROC-based discriminative performance across data splits

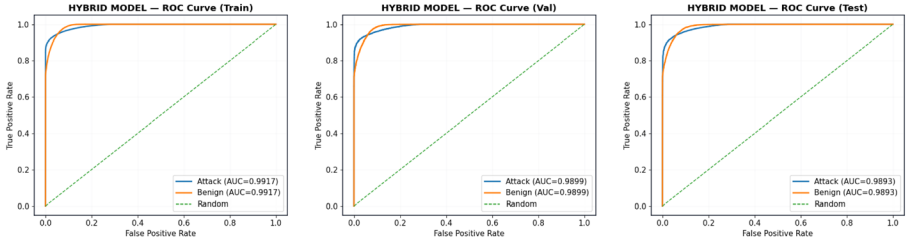
Receiver Operating Characteristic (ROC) curves were used to evaluate the ability of ZD-HybridNet to distinguish attack traffic from benign traffic across the training, validation, and test subsets over different decision thresholds. As shown in Fig. 11a-c, the model achieved AUC values of 0.9917, 0.9899, and 0.9893 on the training, validation, and test subsets, respectively. The small reduction from training to test indicates that the model retains strong discriminative performance on unseen samples without a substantial generalization gap.

The consistently high AUC values and the proximity of the curves to the upper-left region demonstrate effective separation between benign and attack traffic across the evaluated data splits. These results reflect threshold-independent classification performance; they do not constitute a direct evaluation under varying attack intensities or deployment conditions.



(a) Training Set (b) Validation Set (c) Testing Set

Fig. 10 Comparison of confusion matrices across different data splits



(a) Training ROC Curve (b) Validation ROC Curve (c) Testing ROC Curve

Fig. 11 ROC analysis across the training, validation, and test subsets for attack and benign classes

Table 7 Component-wise ablation analysis of ZD-HybridNet

Model variant	Accuracy	Precision	Recall	F1-score
Transformer only	0.9236	0.9429	0.9203	0.9363
LSTM only	0.9152	0.9348	0.9109	0.9275
Without isolation forest score	0.9356	0.9624	0.9340	0.9480
Mean pooling instead of attention pooling	0.9308	0.9559	0.9278	0.9416
Fixed averaging instead of adaptive fusion	0.9321	0.9587	0.9304	0.9443
Full ZD-HybridNet	**0.9404**	**0.9676**	**0.9396**	**0.9534**

Bold values indicate the best performance in each metric

4.7 Component-wise ablation analysis

A component-wise ablation analysis was performed to assess the contribution of the Transformer branch, LSTM branch, Isolation Forest score, attention pooling, and adaptive fusion mechanism. All variants were evaluated using the same experimental protocol.

As shown in Table 7, the complete ZD-HybridNet outperforms the strongest standalone Transformer by 1.68 percentage points in accuracy and 1.71 percentage points in F1-score, confirming the benefit of combining global feature interactions with sequential dependencies. Replacing attention pooling with mean pooling produces the largest component-level F1-score reduction of 1.18 percentage points, followed by fixed averaging in place of adaptive fusion with a 0.91-point decline. Removing the Isolation Forest score reduces the F1-score by 0.54 points. These results indicate that the full model performs best because its two complementary branches, learned attention pooling, adaptive fusion, and anomaly-score feature jointly improve the final representation.

4.8 SHAP-based explainability and feature attribution

Figures 12 and 13 present global SHAP-based explanations of the ZD-HybridNet predictions. The summary plot ranks the processed input features according to their overall influence, where positive SHAP values shift the model output toward the attack class and negative values shift it toward the benign class. The point color indicates the relative magnitude of each feature value.

Using the retained feature-index mapping described in 3.1.4, the four highest-ranked processed features were mapped as follows: f28 corresponds to `[feature name]`, f23 to `[feature name]`, f32 to `[feature name]`, and f31 to `[feature name]`. Their wider SHAP distributions indicate that these traffic attributes exert greater influence on the model output, whereas lower-ranked features show values concentrated closer to zero and therefore make comparatively smaller contributions.

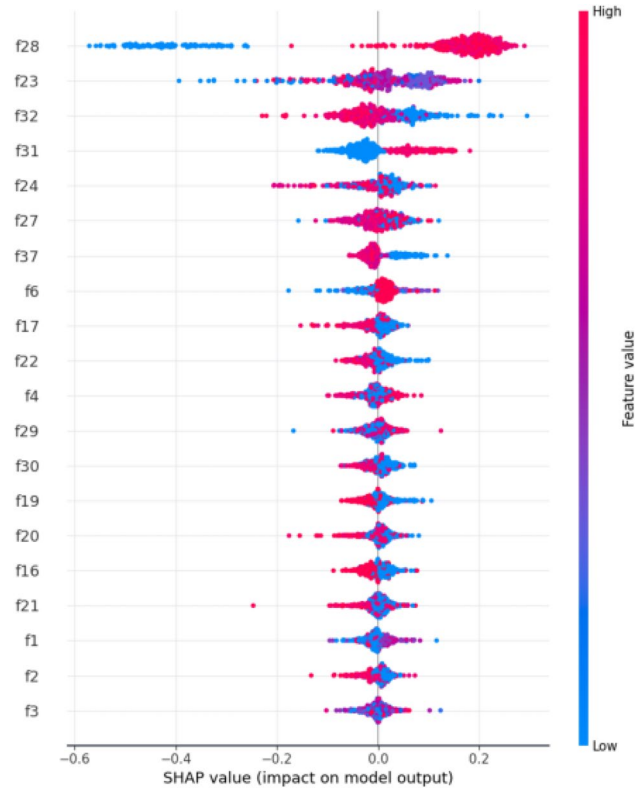


Fig. 12 SHAP-based feature contribution analysis of ZD-HybridNet

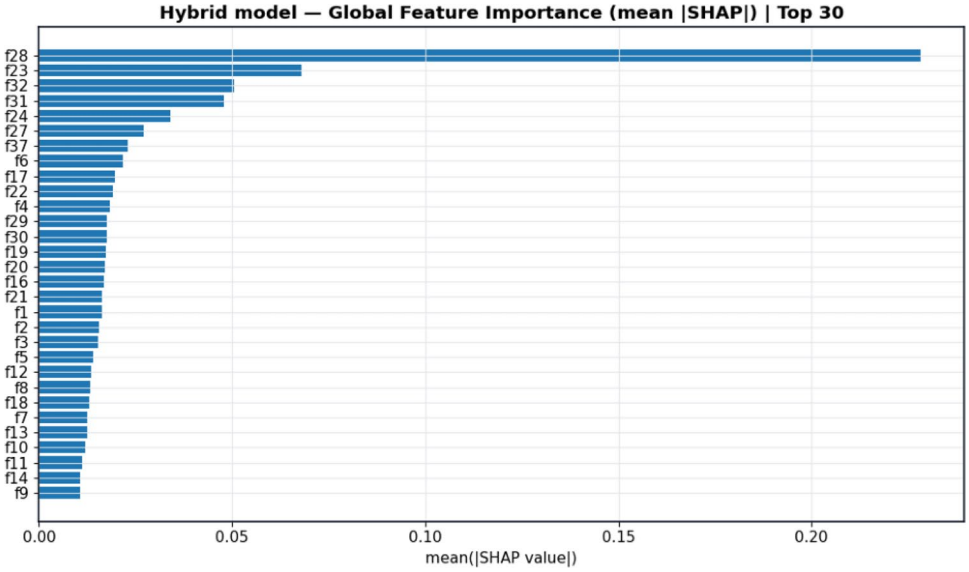


Fig. 13 Global feature importance of ZD-HybridNet based on the mean absolute SHAP values of the top 30 features

The mean absolute SHAP ranking in Fig. 13 confirms that the model relies more strongly on a limited set of discriminative traffic attributes than on the remaining processed features. For security analysts, these ranked contributions can support alert triage by identifying the traffic characteristics that most strongly influenced an attack

Table 8 Contextual comparison with selected zero-day and advanced intrusion detection studies

Refs.	Model/Framework	Reported result and evaluation context
[23]	Attention Fusion IDS	99.99% reported binary accuracy with attention-based interpretability
[25]	GNN–RNN–Attention IDS	97.5% reported accuracy using spatial–temporal representation learning
[41]	ZeroDay-LLM	97.8% reported accuracy with reduced false-positive detection
[28]	ZDBERTa	99.31% reported accuracy using zero-shot learning and GAN-based augmentation
This study	ZD-HybridNet	94.04% accuracy, 96.76% precision, 93.96% recall, and 95.34% F1-score under the Shellcode–Worms category-holdout protocol

prediction. SHAP values are interpreted as post-hoc feature attributions and not as causal evidence of malicious behavior.

4.9 Computational scope and deployment considerations

The experiments were conducted on an NVIDIA Tesla T4 GPU under the computational environment described in 4.1. The present evaluation focuses on detection effectiveness and zero-day generalization rather than deployment-level benchmarking. Therefore, the reported results should not be interpreted as direct evidence of real-time operational performance. Quantitative assessment of inference latency, throughput, model memory consumption, and explanation generation time remains necessary before deployment in latency-sensitive Security Operations Center environments.

4.10 Contextual comparison with existing zero-day detection studies

Table 8 presents a contextual comparison of selected zero-day intrusion detection studies based on their published results. Several referenced approaches report higher accuracy than ZD-HybridNet. However, these values should not be interpreted as a direct performance ranking because the studies employ different datasets, task definitions, data-augmentation strategies, unseen-attack selections, and evaluation protocols.

The recent zero-day methods were therefore included as literature-level references rather than direct experimental baselines. The controlled head-to-head evaluation in this study was conducted using LSTM, Transformer, FT-Transformer and 1D CNN models under an identical preprocessing and partitioning protocol. The intended contribution of ZD-HybridNet is not the highest reported accuracy across heterogeneous studies, but the integration of Transformer–LSTM representations, adaptive fusion, category-isolated zero-day evaluation, and SHAP-based feature attribution within a unified framework.

5 Conclusion

This paper presented ZD-HybridNet, an explainable hybrid deep learning framework designed to address the dual challenges of detecting zero-day cyber attacks and providing interpretable decision rationales in large-scale network environments. By integrating Transformer-based attention mechanisms with LSTM temporal modeling through an adaptive fusion gate, the model effectively captures both global contextual dependencies and localized sequential patterns inherent in network traffic. The explainability framework combines model-internal attention and fusion signals with post-hoc SHAP feature attribution, providing interpretable evidence for examining detection decisions without claiming causal or real-time operational explanations.

Under a controlled category-holdout zero-day evaluation protocol using the UNSW-NB15 dataset, with Shellcode and Worms categorically excluded from training and validation, ZD-HybridNet achieved an accuracy of 94.04%, a precision of 96.76%, a recall of 93.96%, and an F1-score of 95.34%, outperforming the evaluated LSTM, Transformer, FT-Transformer, and 1D CNN baselines under the same experimental protocol. Comprehensive analyses, including precision-recall curves, ROC-AUC evaluation, confusion matrices, zero-day-specific results, ablation analysis, and SHAP-based feature importance interpretation, supported the model's detection and generalization performance within the defined Shellcode–Worms holdout setting. The resulting explanatory signals may assist security analysts in interpreting and prioritizing intrusion alerts, although operational deployment requires further computational validation.

Future work will focus on extending ZD-HybridNet to federated and edge-computing environments to enhance scalability and privacy preservation in distributed network settings. Additionally, exploring continual learning mechanisms to adapt to evolving attack patterns and integrating multi-modal data sources (e.g., logs, packets, flows) represent promising directions for building next-generation, adaptive, and explainable intrusion detection systems. Further evaluation will also consider additional unseen-category holdout scenarios, similarity-aware and session- or time-grouped partitioning, repeated-run statistical analysis, local explanations of representative zero-day cases, visualization of fusion coefficients, and quantitative benchmarking of inference latency, model size, throughput, and memory consumption.

Author contributions

Author Contributions: Conceptualization, R.I. and A.M.; methodology, S.S.J. and K.Y.L.; software, F.M.; validation, M.M. and S.M.R.; formal analysis, V.S.; investigation, A.M. and R.I.; resources, F.M.; data curation, K.Y.L. and S.S.J.; writing—original draft preparation, A.M. and K.Y.L.; writing—review and editing, S.M.R., M.M., and V.S.; visualization, F.M.; supervision, R.I.; project administration, R.I. All authors have read and agreed to the published version of the manuscript.

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Data availability

The UNSW-NB15 dataset analyzed in this study is publicly available through the UNSW Research repository at https://research.unsw.edu.au/projects/unsw-nb15-dataset and the Kaggle repository at https://www.kaggle.com/datasets/mrwellsdavid/unsw-nb15. No new primary dataset was generated during this study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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